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Concept Maps and Semantic Networks
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Concept Maps are mind tools that use knowledge representation schemes with nodes and links between nodes—object, link, label, graphic notation, and others—to represent knowledge. The nodes represent objects and the links represent relations between nodes. This is a concept of semantic networks that is commonly used in cognitive psychology (pg. 104). The mapping approach is also widely used in problem-solving and practice across most professional fields, especially in software and systems engineering, for modeling the software architecture of a system or for problem-solving. More importantly, the concept of mapping can be used for both teaching and learning. For teaching, teachers and educators can use concept maps to represent the learning process in a real-world view of the subject matter and in a natural way people think. For learning, students can learn how to model and to organize the thinking process. In other words, students apply problem-solving by organizing knowledge and sequencing ideas.

A key concept of concept maps is that they model an idea, concept, or problem as an object and its relationships with other objects. This is not only a concept of semantic networks, but databases, spreadsheets, and other modeling tools also use the same technique to organize thought processes. Essentially, it is a mind-mapping technique that stimulates knowledge structures. Semantic networks are also known as cognitive structures, conceptual knowledge, and structural knowledge (pg. 101). The key word here is ‘structure’. Structure represents the world as systems and the natural way people learn and process information. This concept works very similarly to modeling with databases

and spreadsheets, where objects and links between nodes are essentially tables and fields. In fact, databases and spreadsheets are excellent tools to build concept maps.

I have used many computer-based semantic networking software applications to assist me in software design and system architecture. Among my favorite tools are Rational modeling software, Visio, Erwin, and System Architect by Teleologic, among others. These tools are meant for engineers and not for a typical user. I have also used Microsoft Excel, Microsoft Access, and Microsoft Project to gather requirements, brainstorm, and manage projects. Microsoft Excel is an excellent tool to represent a concept or idea and related concepts in a single data grid. Microsoft Access has good facilities for users to create tables, fields, and relationships between tables. Microsoft Project is a good project management tool, and it offers useful graphical tools for creating tasks or breaking a task into smaller sub-tasks. All these tools are classified as data modeling tools, and they are essentially based on the concept of semantic networks.

Knowing how to use computer-based tools is crucial for the successful implementation of any project. On most of my software projects, I used modeling tools to organize my thought processes. A diagram on page 103 (i.e., Figure 10-2) is a good representation of what I have to work with. A user of an application such as Blackboard may not know what is actually happening behind the scenes. When a user clicks on 'User Login', he/she invokes an HTTP request to a remote server (i.e., a web server) somewhere at Alliant, and his/her user ID and password, the authentication information, are sent to the server at Alliant. Many things need to happen before a user receives a response via the Internet browser. One action is to execute a database query to validate the user ID and

password—the authentication process. This is a simple scenario, but if you look at Blackboard as a very complex online application.

I teach computer programming at Ohlone College. Many programming languages, such as .NET, C++, and Java, are object-oriented. An object is simply a representation of the class instance, which is essentially a function, or an object that wraps a function. A typical program has one or more classes. Object-oriented programming also has a concept of an inheritance relationship between two classes. Even though object-oriented is a technical term, it uses many of the same techniques as concept maps and semantic networks.

My students are already familiar with concept maps. They have successfully designed many interesting Web and Windows programming assignments. They have been exposed to database design, data modeling, and class modeling. Finally, they have translated the programming problem into classes and objects. In the .NET IDE (Integrated Development Environment), they have used many built-in features — class, class diagrams and their relationships, and relational database modeling.

Finally, I believe that many people may not realize that, in everyday and workplace life, they have used concept maps for problem-solving. Concept maps and semantic networks are just concepts, but they have been around for years. In addition, I believe that concept maps are a learning skill, and anyone can be taught to master them. I believe that the key to understanding concept maps is the word ‘structure’.